Aashish Kumar Sahu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.50
Chinmaya Vidayalaya, Bokaro	Class XII (CBSE)	2022	88.80%
D.A.V. Public School, Barora	Class X (CBSE)	2020	88.80%

WORK EXPERIENCE

Software Engineering Intern | Kotak Mahindra Bank

KDAC-UI:

- Developed 4 features on KDAC-UI, an enterprise treasury/FX dealing dashboard, using React, TypeScript, Next.js, Ant Design, and TanStack Query.
- Built a strip-conversion flow enabling a single draft deal to split into multiple tenors in one submission, including holiday-calendar-aware date logic.
- Integrated saved RFQ drafts with live market pricing to reduce the manual works of merchants.
- built a CCIL reconciliation page (CSV export/upload) for matching deal records against CCIL data.
- Extended the FX swap deal-booking flow with a new near-leg/far-leg display and custom data-fetching hook.

Founder's Office | PicaPool

- Architected PicaPool's core system design that till works as the basis of PicaPool.
- Studied the entire payment gateway system to integrate an escrow account for secure transactions.

PROJECTS

Sezel -- Persistent AI Assistant

- Designed an async, event-driven cognitive architecture (event bus + finite state machine) for a conversational AI agent combining a local LLM (Ollama) and Claude API, with complexity-based routing between them
- Built a three-tier memory system -- working, episodic (SQLite), and semantic (ChromaDB vector store) -- with a background consolidation process that converts raw conversation history into long-term semantic memory
- Implemented a Valence-Arousal-Dominance affective state model with time-based decay, giving the assistant a persistent, evolving "mood" across sessions
- **Stack:** Python, asyncio, Anthropic API, Ollama, ChromaDB, sentence-transformers, SQLite

Optimal Task Allocation

- Implemented the Hungarian Algorithm ($O(n^3)$) from scratch in C++ to solve a weighted bipartite assignment problem, matching students to tasks by blending assignment cost with a reliability-difficulty compatibility score (user-tunable weighting)
- Built a REST API wrapping the C++ solver and a React/Next.js frontend for interactive data entry and result visualization
- **Stack:** C++ (core algorithm), React, Next.js, Axios, TanStack Query.

SKILLS

Programming Languages: C++, C, SQL, Python, HTML, Javascript, Typescript.

Libraries/Framework: React, Axios, TanStack Query, Next Js, Ant Design

Utilities: Docker, Git, Github, Azure, Kaggle

Data & ML: TensorFlow, Scikit-Learn.

Tools: Vs Code, Anaconda, Jupyter Notebook, Google-Collab.

CS Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, OOPs

POSITION OF RESPONSIBILITY

Placement Coordinator at Office of Career Services

- Pitched 50+ recruiters as internship coordinator and 30+ as Placement Coordinator for the current internship period going on, ensuring seamless onboarding and guiding students for all of their internship doubts.
- Coordinated Placement & Internship Drives 2024 and Japan Day 2024 by managing large student crowd, logistics, guest communication, and on-site operations, delivering flawlessly executed events with high participation and positive feedback.

RELEVANT COURSES

Computer/ AI :- Fundamentals of Scientific Computing, Computational Methods in Materials Science, Introduction to Artificial Intelligence, Introduction to Programming.

Mathematics :- Differential Equation, Calculus - I, Calculus - II, Linear Algebra, Math for Physics.

Physics :- Physics Lab, Modern Physics, Physics of Solid.

L.A./ C.A. :- Communication skills, Ethics and Values, Science and Technology for society, Fundamentals of Organizational Structure

PROGAMMING

Solved 400+ algorithmic problems on platforms like LeetCode and GeeksforGeeks, focusing on data structures (aashish_krshu - LeetCode Profile)

HOBBIES

I enjoy playing Volleyball, cards. They provide great way to connect. I also like singing